DNS SPOOFING ATTACK & FORENSIC ANALYSIS LAB REPORT

Student Name: Leslie Nii Adjetey

Student ID: 2025/CFP/6957

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Instructor: MR. AMINU IDRIS, AMCPN

Executive Summary

This report documents a comprehensive DNS spoofing attack conducted in a controlled laboratory environment as part of the DFIA102 coursework. The exercise successfully demonstrated a complete attack chain involving ARP cache poisoning, Man-in-the-Middle (MitM) positioning, DNS query interception, and HTTP traffic redirection to a fraudulent banking website. The attack was executed using Ettercap for ARP poisoning and DNS spoofing, Apache web server for hosting the phishing page, and Wireshark for comprehensive traffic capture and forensic analysis.

The victim machine (192.168.100.89) was successfully compromised, with all internet-bound traffic redirected through the attacker's Kali Linux machine (192.168.100.6). When the victim attempted to access the intentionally mistyped domain "ww.legitbank.com," the spoofed DNS response directed the browser to the attacker's fake banking login page instead of returning an error or reaching a legitimate site. Complete forensic artifacts were captured, including malicious ARP replies, spoofed DNS responses, and HTTP traffic to the fraudulent server.

This hands-on exercise provided practical insights into network-based attacks and the critical importance of network monitoring, DNSSEC implementation, and the use of HTTPS to detect and prevent such attacks in real-world environments.

Lab Objectives

1. Understand and explain the operational principles of ARP and DNS protocols and how they can be exploited for malicious purposes.
2. Configure an Apache web server to host a convincing phishing page designed to mimic a legitimate banking website.
3. Execute an ARP poisoning attack to establish a Man-in-the-Middle position within a Local Area Network.
4. Perform DNS spoofing to intercept DNS queries and redirect victims to fraudulent websites.
5. Capture comprehensive network traffic using Wireshark during the entire attack lifecycle.
6. Analyze captured network traffic to identify and document forensic evidence of ARP poisoning and DNS spoofing.
7. Compile a professional forensic report detailing attack methodology, findings, and indicators of compromise.

Lab Environment and Network Configuration

Attacker Machine (Kali Linux)

IP Address: 192.168.100.6
MAC Address: 08:00:27:38:2a:25 (Oracle VirtualBox virtual NIC)
Interface: eth0
Operating System: Kali Linux
Role: Attack orchestration, MitM position, fake web server hosting

Victim Machine (Ubuntu)

IP Address: 192.168.100.89
MAC Address: 08:00:27:02:c7:dd (Oracle VirtualBox virtual NIC)
Interface: enp0s3
Operating System: Ubuntu Linux
Role: Target of ARP poisoning and DNS spoofing

Network Gateway/Router

IP Address: 192.168.100.1
MAC Address: a0:40:6f:59:11:83 (Huawei Technologies)
Role: Default gateway for the subnet

Network Subnet

Network Range: 192.168.100.0/24
Subnet Mask: 255.255.255.0
Broadcast Address: 192.168.100.255

Tools and Resources Used

Ettercap 0.8.3.1 - For ARP poisoning and DNS spoofing attacks
Wireshark 4.4.7 - For comprehensive packet capture and traffic analysis
Apache2 Web Server - For hosting the fake banking login page
ifconfig/ip - For network interface configuration and verification
arp - For viewing and verifying ARP cache manipulation on victim machine
nano/cat - For creating and managing configuration files
curl - For testing web server accessibility
nmap - For network reconnaissance and host discovery

Methodology

Methodology

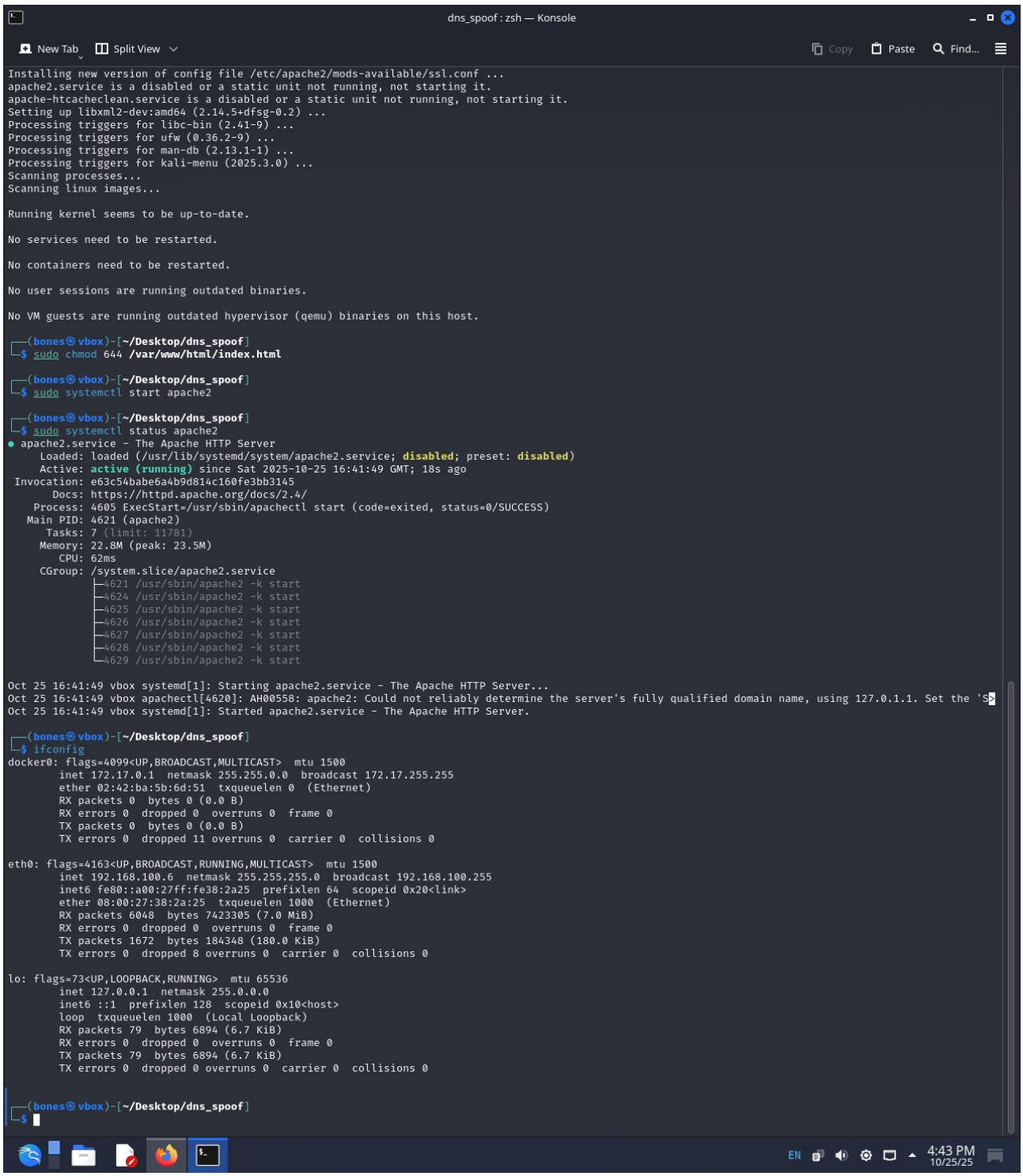
The attack was carried out in phases, following standard steps for setup, execution, and verification.

Environment Setup

I set up the lab using VirtualBox with Kali Linux as the attacker machine (IP: 192.168.100.6, MAC: 08:00:27:38:2a:25) and Ubuntu as the victim (IP: 192.168.100.89, MAC: 08:00:27:02:c7:dd). The

gateway was at 192.168.100.1 (MAC: a0:40:6f:59:11:83). I created a working directory on Kali, installed and configured Apache to host a fake banking page at /var/www/html/index.html. The page mimicked a bank login with fields for username and password. I verified it by accessing http://192.168.100.6 using curl.

```
dns_spoof : zsh — Konsole
New Tab   Split View                                                    Copy   Paste   Find...

        <label for="username">Username:</label>
        <input type="text" id="username" name="username" required><br><br>

        <label for="password">Password:</label>
        <input type="password" id="password" name="password" required><br><br>

        <input type="submit" value="Login">
    </form>
</body>

</html>

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ sudo cp index.html /var/www/html/.
[sudo] password for bones:

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ ll /var/www/html/index.html
-rw-r--r-- 1 root root 469 Oct 25 15:42 /var/www/html/index.html

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ sudo apt install apache2 -y
[sudo] password for bones:
The following packages were automatically installed and are no longer required:
  apg                          libgail18t64                 libkf5khtml-data           libphonon4qt5-4t64                  python3-pyinstaller-hooks-contrib
  fonts-inter-variable         libgeos3.13.0                libkf5khtml5               libpoppler145                       python3-pywerview
  gnome-accessibility-themes   libgit2-1.8                  libkf5package-data         libpython3.12-minimal               python3-requests-ntlm
  gnome-themes-extra           libglapi-mesa                libkf5package5             libpython3.12-stdlib                python3-setproctitle
  hwdata                       libgrantlee-templates5       libkf5parts-data           libpython3.12t64                    python3-tomlkit
  icu-devtools                 libgtk2.0-0t64               libkf5parts-plugins        libutempter0                        python3-wheel-whl
  kde-style-oxygen-qt6         libgtk2.0-bin                libkf5parts5               libxnnpack0                         python3.12-tk
  kdoctools5                   libgtk2.0-common             libkf5plasma5              phonon4qt5                          qml6-module-qtqml-base
  kpackagetool5                libicu-dev                   liblbfgsb0                 phonon4qt5-backend-vlc              ruby-zeitwerk
  libabsl20230802              libjxl0.10                   libopencv-core406t64       python3-aioconsole                  sphinx-rtd-theme-common
  libdnnl3                     libkdecorations2private11    libopencv-imgproc406t64    python3-dunamai                     tracker
  libflac12t64                 libkf5idletime5              libopenh264-7              python3-nfsclient                   tracker-miner-fs
  libfuse3-3                   libkf5js5                    liboxygenstyle6-6          python3-packaging-whl
  libgail-common               libkf5khtml-bin              liboxygenstyleconfig6-6    python3-poetry-dynamic-versioning
Use 'sudo apt autoremove' to remove them.

Upgrading:
  apache2  apache2-bin  apache2-data  apache2-utils  libxml2-dev  libxml2-utils

Installing dependencies:
  liblzma-dev  libxml2-16

Suggested packages:
  liblzma-doc

Summary:
  Upgrading: 6, Installing: 2, Removing: 0, Not Upgrading: 1185
  Download size: 3,859 kB
  Space needed: 2,092 kB / 16.6 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 libxml2-16 amd64 2.14.5+dfsg-0.2 [638 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 apache2 amd64 2.4.65-3+b1 [226 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 apache2-bin amd64 2.4.65-3+b1 [1,405 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 apache2-data all 2.4.65-3 [160 kB]
Get:5 http://http.kali.org/kali kali-rolling/main amd64 apache2-utils amd64 2.4.65-3+b1 [216 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 liblzma-dev amd64 5.8.1-1 [348 kB]
Get:7 http://http.kali.org/kali kali-rolling/main amd64 libxml2-dev amd64 2.14.5+dfsg-0.2 [744 kB]
Get:8 http://http.kali.org/kali kali-rolling/main amd64 libxml2-utils amd64 2.14.5+dfsg-0.2 [123 kB]
Fetched 3,859 kB in 1s (4,496 kB/s)
Selecting previously unselected package libxml2-16:amd64.
(Reading database ... 552656 files and directories currently installed.)
Preparing to unpack .../0-libxml2-16_2.14.5+dfsg-0.2_amd64.deb ...
Unpacking libxml2-16:amd64 (2.14.5+dfsg-0.2) ...
Preparing to unpack .../1-apache2_2.4.65-3+b1_amd64.deb ...
Unpacking apache2 (2.4.65-3+b1) over (2.4.63-1) ...
Preparing to unpack .../2-apache2-bin_2.4.65-3+b1_amd64.deb ...
Unpacking apache2-bin (2.4.65-3+b1) over (2.4.63-1) ...
Preparing to unpack .../3-apache2-data_2.4.65-3_all.deb ...
Unpacking apache2-data (2.4.65-3) over (2.4.63-1) ...
Preparing to unpack .../4-apache2-utils_2.4.65-3+b1_amd64.deb ...
Unpacking apache2-utils (2.4.65-3+b1) over (2.4.63-1) ...
Selecting previously unselected package liblzma-dev:amd64.
Preparing to unpack .../5-liblzma-dev_5.8.1-1_amd64.deb ...
Unpacking liblzma-dev:amd64 (5.8.1-1) ...
Preparing to unpack .../6-libxml2-dev_2.14.5+dfsg-0.2_amd64.deb ...
Unpacking libxml2-dev:amd64 (2.14.5+dfsg-0.2) over (2.12.7+dfsg+really2.9.14-1) ...
Preparing to unpack .../7-libxml2-utils_2.14.5+dfsg-0.2_amd64.deb ...
Unpacking libxml2-utils (2.14.5+dfsg-0.2) over (2.12.7+dfsg+really2.9.14-1) ...
Setting up libxml2-16:amd64 (2.14.5+dfsg-0.2) ...
EN   4:43 PM 10/25/25
```



Attack Execution

First, I enabled IP forwarding on Kali with echo 1 > /proc/sys/net/ipv4/ip_forward to keep the victim's internet working. Then, I configured DNS spoofing in Ettercap by editing /etc/ettercap/etter.dns to map domains like ww.legitbank.com to 192.168.100.6. I launched Ettercap with sudo ettercap -T -q -i eth0 -M arp:remote /192.168.100.89// /192.168.100.1// -P dns_spoof to poison ARP and spoof DNS. This positioned the attacker as a Man-in-the-Middle.

dns_spoof : zsh — Konsole

New Tab Split View Copy Paste Find...

```
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ sudo chmod 644 /var/www/html/index.html

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ sudo systemctl start apache2

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
     Active: active (running) since Sat 2025-10-25 16:41:49 GMT; 18s ago
 Invocation: e63c54babe6a4b9d814c160fe3bb3145
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 4605 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 4621 (apache2)
      Tasks: 7 (limit: 11781)
     Memory: 22.8M (peak: 23.5M)
        CPU: 62ms
     CGroup: /system.slice/apache2.service
             ├─4621 /usr/sbin/apache2 -k start
             ├─4624 /usr/sbin/apache2 -k start
             ├─4625 /usr/sbin/apache2 -k start
             ├─4626 /usr/sbin/apache2 -k start
             ├─4627 /usr/sbin/apache2 -k start
             ├─4628 /usr/sbin/apache2 -k start
             └─4629 /usr/sbin/apache2 -k start

Oct 25 16:41:49 vbox systemd[1]: Starting apache2.service - The Apache HTTP Server...
Oct 25 16:41:49 vbox apachectl[4620]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'S>
Oct 25 16:41:49 vbox systemd[1]: Started apache2.service - The Apache HTTP Server.

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        ether 02:42:ba:5b:6d:51  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 11 overruns 0  carrier 0  collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.100.6  netmask 255.255.255.0  broadcast 192.168.100.255
        inet6 fe80::a00:27ff:fe38:2a25  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:38:2a:25  txqueuelen 1000  (Ethernet)
        RX packets 6048  bytes 7423305 (7.0 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1672  bytes 184348 (180.0 KiB)
        TX errors 0  dropped 8 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 79  bytes 6894 (6.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 79  bytes 6894 (6.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0


┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward

1

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ cat /proc/sys/net/ipv4/ip_forward
1

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$
```

EN 4:44 PM

Attack Verification

From the victim, I checked the ARP cache before and after. Initially, it showed the correct gateway MAC; after poisoning, it showed the attacker's MAC for the gateway IP. I then browsed to http://ww.legitbank.com in Firefox, and it loaded the fake page hosted on the attacker.

Throughout, I captured traffic on Kali's eth0 using Wireshark.

Traffic Analysis

The analysis focused on key protocols using Wireshark filters to isolate evidence.

ARP Poisoning Evidence

Using filter arp.opcode == 2 && arp.src.proto_ipv4 == 192.168.100.1, I found packets where the attacker's MAC claimed the gateway's IP. For example, packet 64 showed: Sender MAC: 08:00:27:38:2a:25 (attacker), Sender IP: 192.168.100.1 (gateway, spoofed), Target MAC: 08:00:27:02:c7:dd (victim), Target IP: 192.168.100.89. This is a gratuitous ARP reply that poisons the cache, allowing traffic interception. Multiple such packets appeared, maintaining the poisoning.

dns_spoof_attack_sim.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp

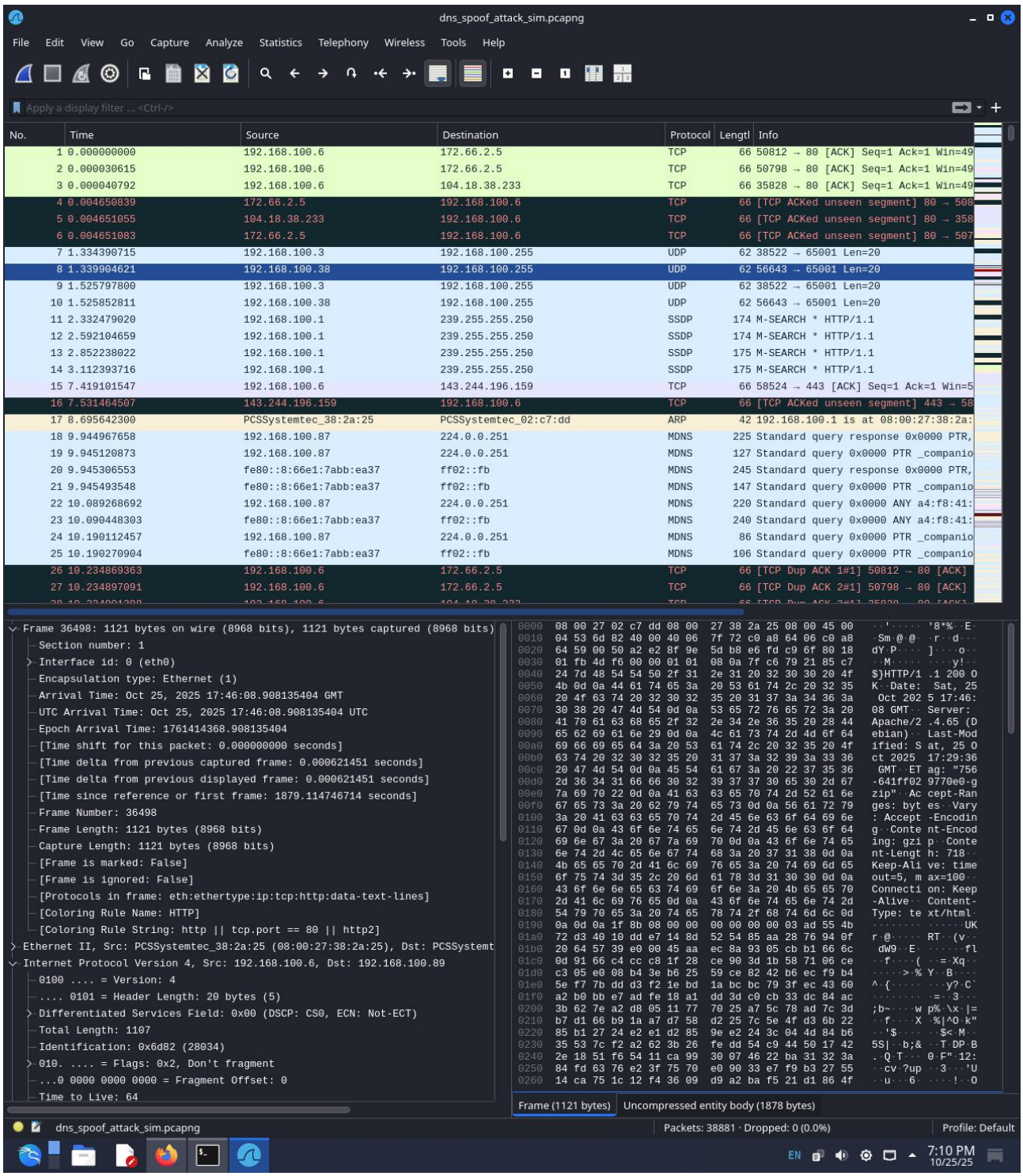
No.	Time	Source	Destination	Protocol	Length	Info
17	8.695642300	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
38	10.695869966	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
47	12.552149937	PCSSystemtec_38:2a:25	HuaweiTechno_59:11:83	ARP	42	Who has 192.168.100.1? Tell 192.168
48	12.552655703	HuaweiTechno_59:11:83	PCSSystemtec_38:2a:25	ARP	60	192.168.100.1 is at a0:40:6f:59:11:
49	12.712014252	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
57	14.716340916	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
64	16.718737970	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
72	18.718988686	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
82	20.719590387	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
85	22.719811439	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
86	24.726617062	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
87	26.727302524	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
90	28.426984151	PCSSystemtec_38:2a:25	HuaweiTechno_59:11:83	ARP	42	192.168.100.89 is at 08:00:27:38:2a
91	28.440067180	HuaweiTechno_59:11:83	Broadcast	ARP	60	Who has 192.168.100.89? Tell 192.16
96	28.727564124	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
97	30.427237177	PCSSystemtec_38:2a:25	HuaweiTechno_59:11:83	ARP	42	192.168.100.89 is at 08:00:27:38:2a
104	30.728803443	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
105	31.774562921	HuaweiTechno_59:11:83	PCSSystemtec_38:2a:25	ARP	60	Who has 192.168.100.89? Tell 192.16
133	32.431274505	PCSSystemtec_38:2a:25	HuaweiTechno_59:11:83	ARP	42	192.168.100.89 is at 08:00:27:38:2a
134	32.557899836	HuaweiTechno_59:11:83	Broadcast	ARP	60	Who has 192.168.100.87? Tell 192.16
135	32.729124403	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
140	33.631185443	HuaweiTechno_59:11:83	Broadcast	ARP	60	Who has 192.168.100.87? Tell 192.16
141	34.433473935	PCSSystemtec_38:2a:25	HuaweiTechno_59:11:83	ARP	42	192.168.100.89 is at 08:00:27:38:2a
142	34.671031949	HuaweiTechno_59:11:83	Broadcast	ARP	60	Who has 192.168.100.87? Tell 192.16
143	34.730940517	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:
144	36.434552158	PCSSystemtec_38:2a:25	HuaweiTechno_59:11:83	ARP	42	192.168.100.89 is at 08:00:27:38:2a
145	36.738852642	PCSSystemtec_38:2a:25	PCSSystemtec_02:c7:dd	ARP	42	192.168.100.1 is at 08:00:27:38:2a:

```
Frame 64: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interfac
    Section number: 1
    Interface id: 0 (eth0)
    Encapsulation type: Ethernet (1)
    Arrival Time: Oct 25, 2025 17:15:06.512126660 GMT
    UTC Arrival Time: Oct 25, 2025 17:15:06.512126660 UTC
    Epoch Arrival Time: 1761412506.512126660
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.003281123 seconds]
    [Time delta from previous displayed frame: 2.002397054 seconds]
    [Time since reference or first frame: 16.718737970 seconds]
    Frame Number: 64
    Frame Length: 42 bytes (336 bits)
    Capture Length: 42 bytes (336 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:arp]
    [Coloring Rule Name: ARP]
    [Coloring Rule String: arp]
Ethernet II, Src: PCSSystemtec_38:2a:25 (08:00:27:38:2a:25), Dst: PCSSystemtec_
Address Resolution Protocol (reply)
```

```
0000  08 00 27 02 c7 dd 08 00  27 38 2a 25 08 06 00 01   ··'····· '8*%····
0010  08 00 06 04 00 02 08 00  27 38 2a 25 c0 a8 64 01   ........ '8*%··d·
0020  08 00 27 02 c7 dd c0 a8  64 59                     ··'····· dY
```

Address Resolution Protocol: Protocol | Packets: 38881 · Displayed: 2687 (6.9%) · Dropped: 0 (0.0%) | Profile: Default

EN 5:57 PM 10/25/25



DNS Spoofing Evidence

With filter dns && ip.addr == 192.168.100.89, packet 36481 showed the victim's query for ww.legitbank.com (Type A). The response spoofed it to 192.168.100.6 with TTL 3600. Ettercap logs confirmed: "dns_spoof: A [ww.legitbank.com] spoofed to [192.168.100.6]". This fake response redirects the victim without reaching a real DNS server.

dns_spoof : zsh — Konsole

New Tab Split View Copy Paste Find...

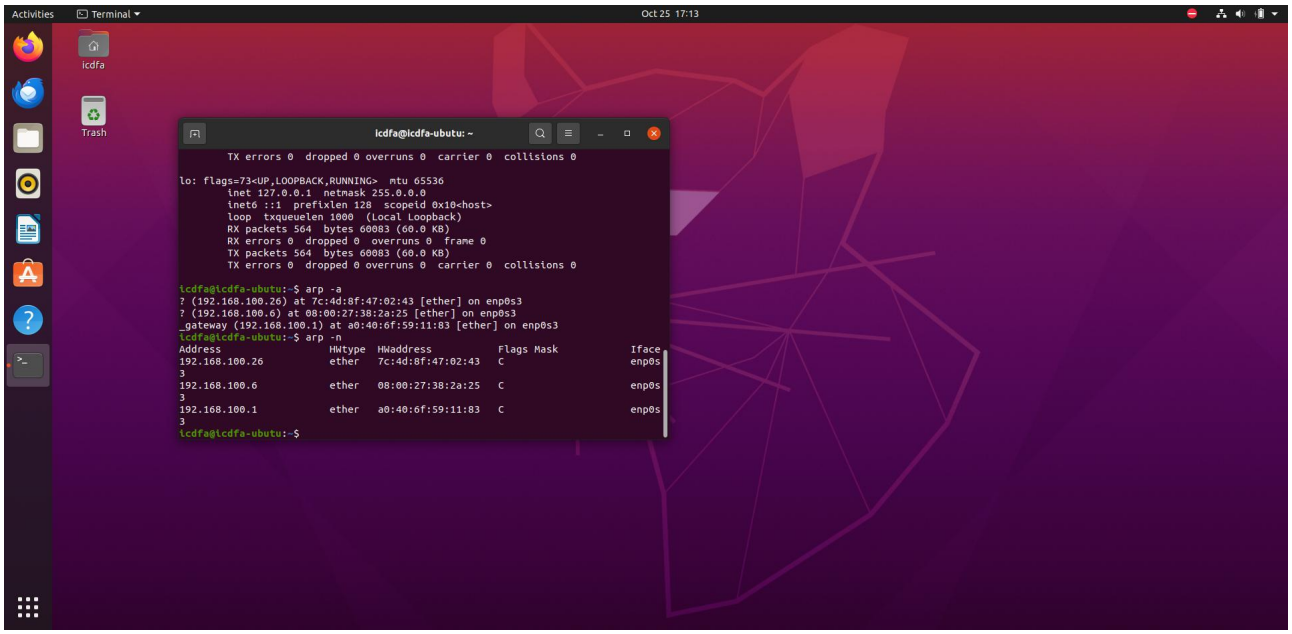
-: sudo -: sudo dns_spoof : zsh

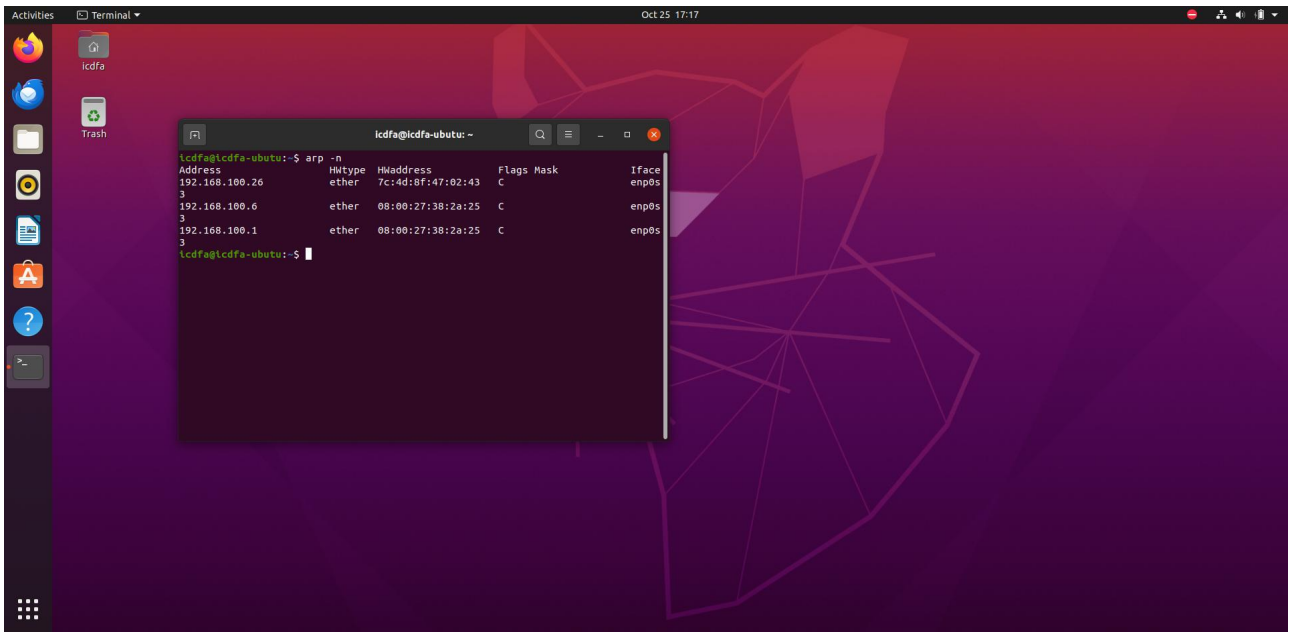
```
┌──(bones㉿vbox)-[~]
└─$ cd Desktop

┌──(bones㉿vbox)-[~/Desktop]
└─$ cd dns_
cd: no such file or directory: dns_

┌──(bones㉿vbox)-[~/Desktop]
└─$ cd dns_spoof

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ echo "192.168.100.6 ww.legitbank.com" > dns_hosts.txt

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ ls
dns_hosts.txt  index.html  index.html.backup

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ echo "192.168.100.6 www.fakebank.com" >> dns_hosts.txt

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ echo "192.168.100.6 secure-bank.com" >> dns_hosts.txt

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ ls\
>

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ ls
dns_hosts.txt  index.html  index.html.backup

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$ cat dns_hosts.txt
192.168.100.6 ww.legitbank.com
192.168.100.6 www.fakebank.com
192.168.100.6 secure-bank.com

┌──(bones㉿vbox)-[~/Desktop/dns_spoof]
└─$
```

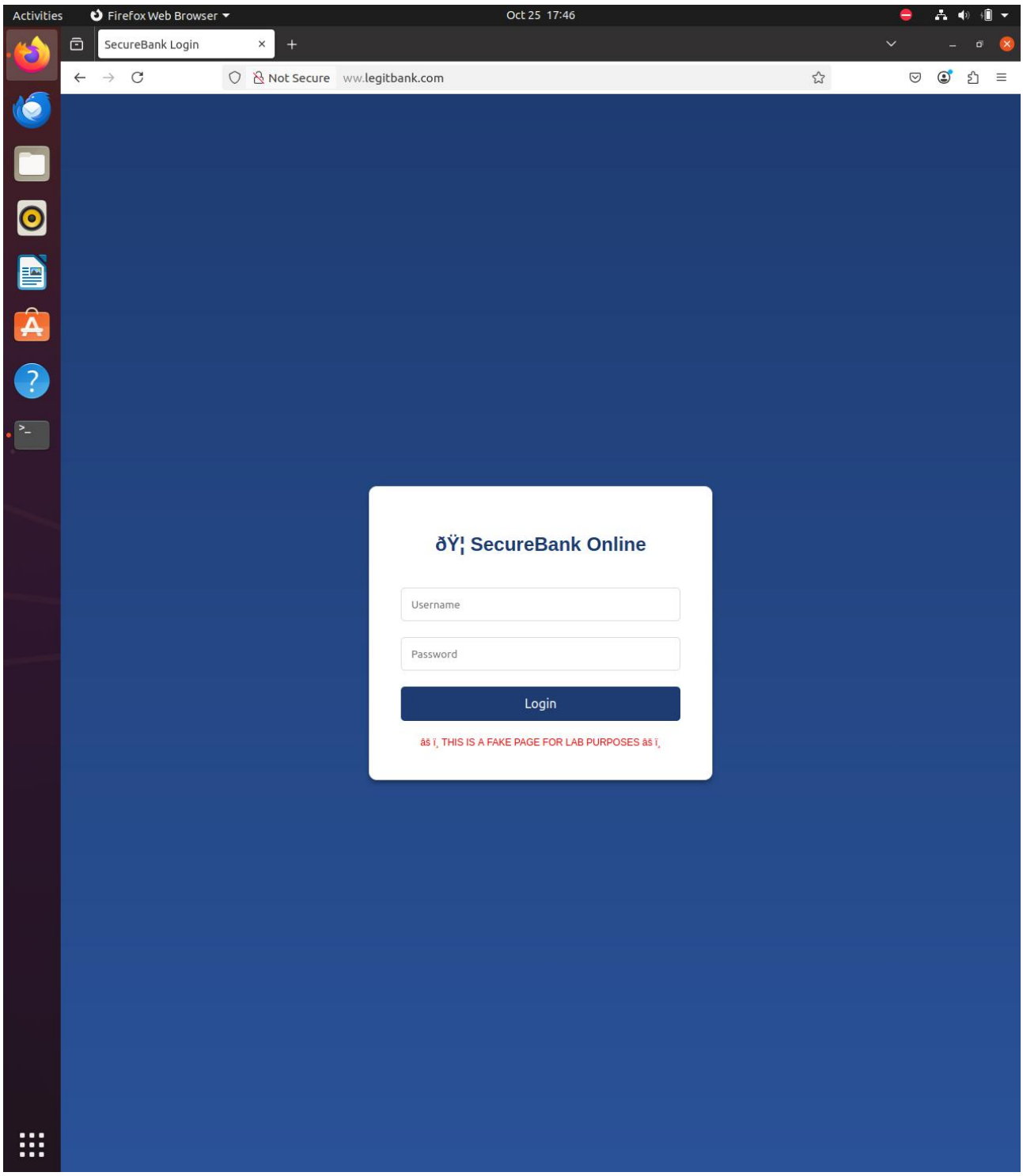
EN 5:32 PM 10/25/25

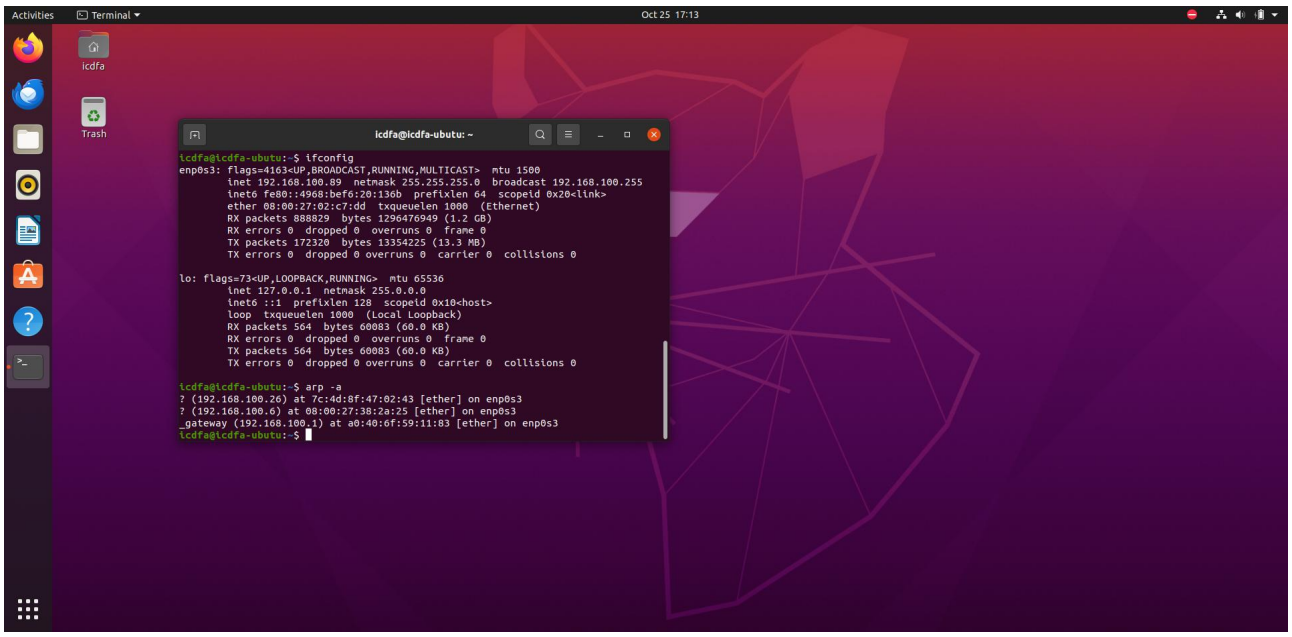
HTTP Redirection

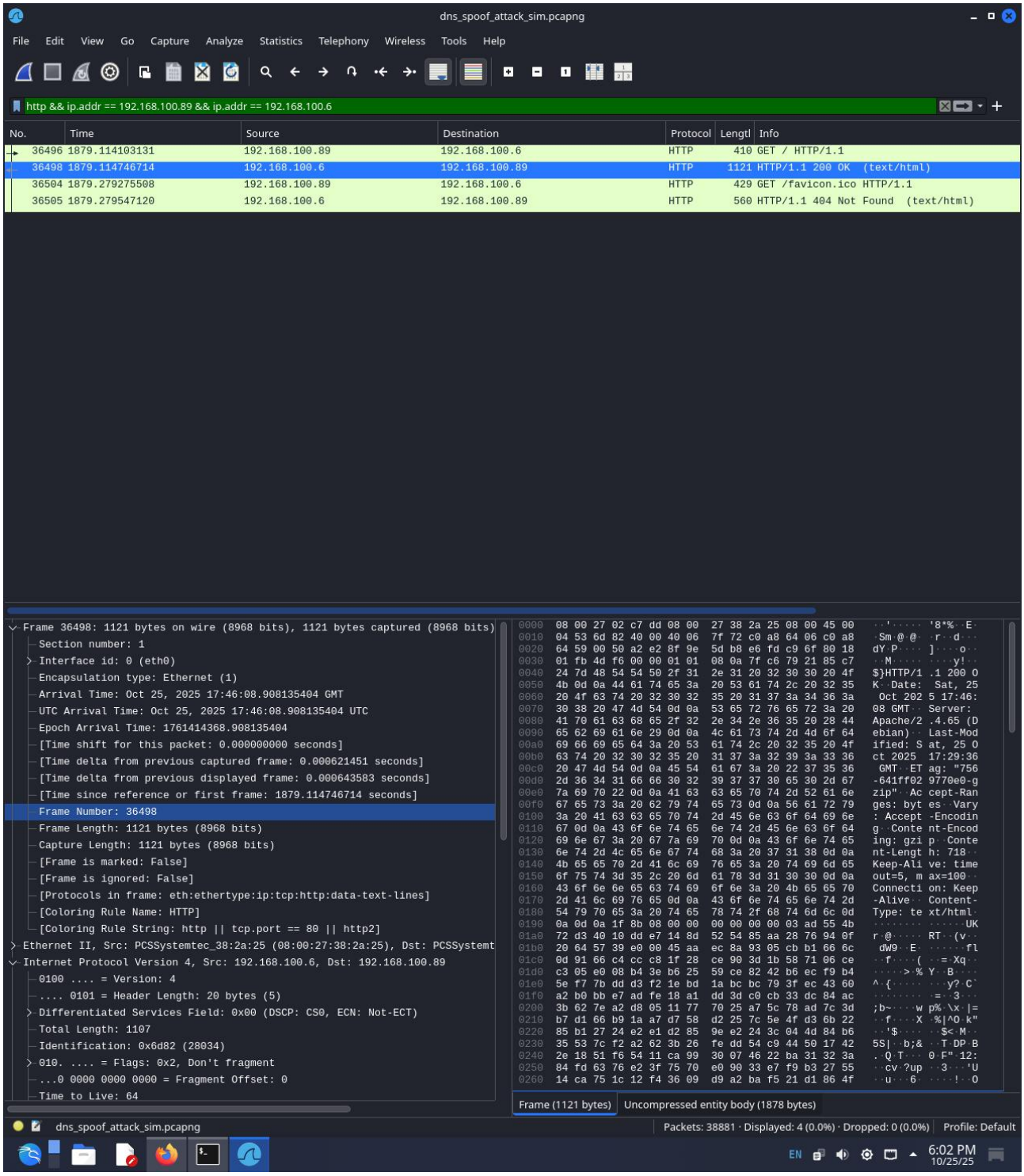
Filter http && ip.addr == 192.168.100.89 && ip.addr == 192.168.100.6 revealed packet 36496: Victim's HTTP GET / from 192.168.100.89 to 192.168.100.6. Response packet 36498: HTTP 200 OK, Content-Type: text/html, Length: 1121 bytes, Server: Apache/2.4.65-1+b1 (Debian). This confirms the redirection led to loading the fake page.






Activities
Firefox Web Browser
Oct 25 17:46
SecureBank Login
Not Secure
www.legitbank.com
ðŸ¦ SecureBank Online
Username
Password
Login
âš ï¸ THIS IS A FAKE PAGE FOR LAB PURPOSES âš ï¸


Activities
Terminal
Oct 25 17:13
icdfa
Trash
icdfa@icdfa-ubutu: ~
icdfa@icdfa-ubutu:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.100.89 netmask 255.255.255.0 broadcast 192.168.100.255
inet6 fe80::4968:bef6:20:136b prefixlen 64 scopeid 0x20<link>
ether 08:00:27:02:c7:dd txqueuelen 1000 (Ethernet)
RX packets 888829 bytes 1296476949 (1.2 GB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 172320 bytes 13354225 (13.3 MB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 564 bytes 60083 (60.0 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 564 bytes 60083 (60.0 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
icdfa@icdfa-ubutu:~$ arp -a
? (192.168.100.26) at 7c:4d:8f:47:02:43 [ether] on enp0s3
? (192.168.100.6) at 08:00:27:38:2a:25 [ether] on enp0s3
_gateway (192.168.100.1) at a0:40:6f:59:11:83 [ether] on enp0s3
icdfa@icdfa-ubutu:~$



Key Findings & Indicators of Compromise (IoCs)

The analysis confirmed the attack through these IoCs:

Attacker machine: IP 192.168.100.6, MAC 08:00:27:38:2a:25 (Kali VM).
Spoofed DNS query/response: Query for ww.legitbank.com from 192.168.100.89, response directing to 192.168.100.6.
Fake IP for redirection: 192.168.100.6.

Other signs include duplicate MACs in ARP cache, gratuitous ARP replies, fast local DNS responses, and HTTP connections to the attacker's server with phishing content.

Conclusion

The attack proved effective in the lab, exploiting ARP and DNS without authentication. It succeeded fully, highlighting risks from local access. Wireshark provided solid evidence for detection. To prevent this, implement network monitoring rules like alerting on gratuitous ARP replies or mismatched MAC-IP mappings, and enforce DNSSEC to validate responses. A layered defense is key.